

Qingyuan YAO

Graduate of the AI master's program at Sorbonne University

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Portfolio	https://mrzny.github.io/
Github	https://github.com/MRVNY

Education

2021 - 2023	Master of Artificial Intelligence <i>Parcours AgeNts Distribués, Robotique, Recherche Opérationnelle, Interaction, DEcision (ANDROIDE)</i> Mention Bien Sorbonne University
2018 - 2021	Bachelor of Computer Science Mention Bien Sorbonne University

Experience

2023	Deep Generative AI Internship Tokyo National Institute of Informatics <i>I worked in the domain of dance synthesis, my work consisted of creating a new dataset of audio-text-motion by extracting motion data from videos using existing deep learning models 3D pose estimation, and implementing a <u>Transformer VAE</u> baseline model to showcase the validity of adding lyrics as a condition in the process of dance motion generation.</i>
2022	Serious Game Development Internship Team MOCAH of LIP6 at Sorbonne University <i>I was the solo developer of a serious game that raises awareness of the living conditions of chamois in the Bauges as well as the impact of human activities. The game is set out to show the research result of the project HUMANI at the University of Savoie.</i> <i>The internship is based on the prototype already developed by a previous team of students. During the internship, I hugely improved the stability and optimization of the game, while adding new implementations into the game such as a visual novel interface with loadable json, <u>A* pathfinding</u> system, asynchronous execution, and a more extensive save&load system.</i>
2021 - Now	Game Jams <i>I have participated in the multiple game jams for 3 years, coding in Unity and Unreal5. I have collaborated with programmers and artists in the process (cf. portfolio)</i>
2019 - 2022	Game Design Workshop Sorbonne University <i>Learned the basics of game design and made 2 board-game prototypes (cf. portfolio)</i>
2021 - Now	Chorister at the University Choir La Clef des Chants Ensemble
2019 - 2020	Sorbonne Debate Club Sorbonne University

Skills

- **Game development**
 - Experienced in **Unity**
 - Beginner in **Unreal 5**
- **Multi-Agent System**
- **Human-Computer Interaction Protocols**
- **Deep Learning**
 - Reinforcement Learning
 - Decision, Game Theory and Operational Research
 - Web development
 - Media production using Lightroom, Final Cut, Logic
 - Analog photography

Interests

- Indie Games
- Game Design
- Human-Computer Interaction
- Sociology
- Gender Theory
- Feminism
- Language
- Choir
- Songwriting
- Cinema
- Pop culture
- Analog & Digital Photography

Languages

- **Chinese:** Native
- **English:** C2
- **French:** C1 - C2
- **Spanish:** A2 - B1
- **Japanese:** A1 - A2
- **Python**
- **C & C# & C++**
- **Java & JavaScript**
- **Blueprint**
- **LaTEX & Markdown**

Projects

2023	追忆 (Zhuiyi) <i>Zhuiyi - In Search of Lost Words is a 3D first-person perspective game that teaches people to write 5 simple Chinese characters. The game takes a Wacom graphic tablet with a Nintendo Switch Joycon, or keyboard with mouse as input methods. We have implemented the <u>\$P recognizer</u> for character recognition, and <u>Bayesian Knowledge Tracing (BKT)</u> for player modeling and adaptation.</i>
2023	SPY <i>Enhancement of the serious game SPY that teaches children basic programming notions. Our team added a level editor, a visual novel UI, a map that implements the <u>Competence-based Knowledge Space Theory (CbKST)</u>, as well as a tracing mechanism using <u>xAPI</u>.</i>
2023	Meta-Reinforcement Learning Project <i>Reproduction and analysis of the paper <u>Prefrontal Cortex as a Meta-Reinforcement Learning System</u>, which consists of the adaptation of a <u>LSTM</u> model with <u>A2C</u> in Tensorflow.</i>
2023	Multi-Agent Simulation of Ecotopia <i>Simulation of transport, housing, energy, and electricity in a society based on the book Ecotopia. I worked on the housing section of the development in GAML.</i>
2022	HCI Project: Immersive Book <i>A Human-Computer Interaction project where we designed the concept of a holographic interactive book, with the practice of user-centered design while following HCI protocols such as making interviews, defining interaction points, persona, design dimensions, and in the end prototyping a demo video.</i>
2022	HCI Project: Gestures vs. Keyboard <i>Development of a website that collects and compares gestural and keyboard shortcuts under HCI protocols</i>
2022	Projet Dedale <i>Project of the course Fundamentals of Multi-Agent Systems where agents work together to collect treasures</i>